CoreKD: A Context-Aware Local Region Structural Contrastive Knowledge Distillation Framework for Object Detection

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***Abstract*—Knowledge distillation (KD) aims to transfer knowledge from a cumbersome teacher to a lightweight student, thereby reducing overall model complexity without sacrificing performance. Current methods tend to focus excessively on pixel-level knowledge transfer while overlooking localized and contextual information. To address this, we propose a novel context-aware local region structural contrastive knowledge distillation framework (CoreKD) for object detection tasks. Specifically, we introduce a patch-based semantic structural distillation (PSD) method, facilitating efficient localized valuable knowledge transfer. This method guides the student model in learning both consistency and diverse local knowledge from the teacher model by integrating the semantic and structural information. We also propose the intra-region contextual (Ita-RC) and inter-region contextual (Ite-RC) constraints for the PSD framework. These constraints could transfer the region-wise contextual information between the student and teacher, significantly enhancing the student detector's capacity to model contextual dependencies. To validate the efficacy of CoreKD, we conduct extensive experiments on the public MS COCO 2017 and PASCAL VOC datasets. These results demonstrate that CoreKD markedly improves the performance of students in object detection tasks, indicating its ability to transfer valuable knowledge.**

***Index Terms*—Contextual alignment, knowledge distillation (KD), object detection, semantic structural contrastive.**

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I. Introduction

As a fundamental task in computer vision fields, object detection has found extensive applications in various fields such as autonomous driving [1], industrial inspection [2], and pose estimation [3]. The rapid advancements have significantly propelled the field of object detection forward [4]. Leveraging the powerful representational capabilities of deep learning, current detectors [5], [6], [7], [8] have achieved impressive performance. To fully utilize its representational power, these detectors often adopt heavy backbone networks (e.g., ResNet [9] with 101 layers). However, this results in substantial computational resource requirements, significantly limiting their deployment and practical application in real-world scenarios.

To achieve the deployment of neural network-based detector methods on edge devices, knowledge distillation (KD) [10] is a critical well-established technique for model compression and has been widely applied across various tasks such as classification [11], [12], [13], object detection [14], activity recognition [15], and defect inspection [16]. KD involves incorporating the knowledge from a cumbersome yet well-trained model (teacher) into a lightweight model (student) during the training process, thereby enhancing the student's learning capabilities. In object detection, KD strategies can be broadly classified into feature distillation [17], [18], [19], [20], [21], [22], [23], [24] and logits distillation [25], [26], [27], [28], [29], based on the distillation target [30]. Feature distillation is particularly favored over logit distillation due to its ability to capture rich spatial knowledge.

Most feature imitation methods [18], [19], [20], [23], [29] aim to minimize the distance between teacher and student features. However, we observe several limitations as follows.

1) *Overemphasis on Pixel-Level Knowledge:* While individual pixels lack semantic information, aggregated pixel patterns convey specific semantic content. In object detection, an overemphasis on pixel-level knowledge may cause the student model to learn irrelevant or redundant background information while ignoring valuable local structures. As illustrated in the red boxes of Fig. 1, the teacher model effectively captures wheel-related localized knowledge, whereas the student model fails to do so, leading to incorrect predictions. Transferring such local knowledge could significantly improve object recognition performance.

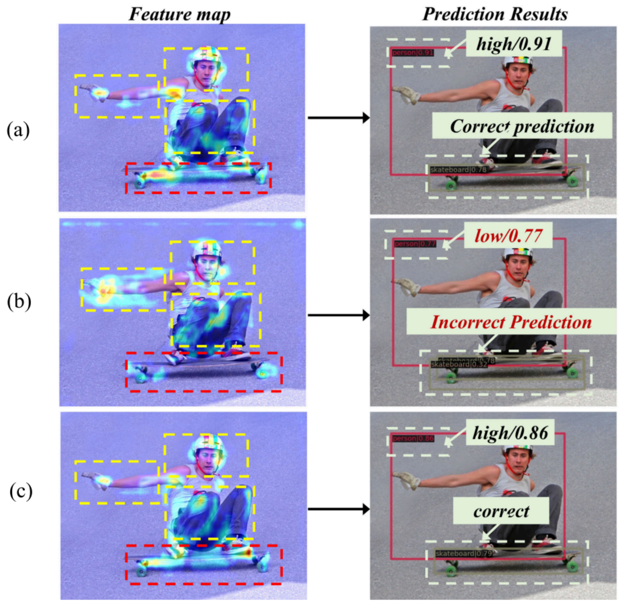

Fig. 1. Comparison of heatmaps and prediction results of teacher and student models. The teacher model is a GFL [7] detector with ResNet-101 [9], and the student model is a GFL [7] detector with ResNet-50 [9]. (a) Teacher. (b) Student without CoreKD. (c) Student with CoreKD.

2) *Neglect of Contextual Knowledge:* As shown in the yellow boxes of Fig. 1(a) and (b), the teacher model simultaneously focuses on local semantic information (e.g., arms, head, and legs) and global semantic relationships (e.g., spatial dependencies between these parts). In contrast, the student model overly concentrates on arm regions, indicating a lack of comprehensive contextual understanding. This results in a significant confidence gap between the teacher and student models (91% versus 77%). Transferring the teacher's ability to model contextual relationships could further enhance the student's performance.

In this article, we propose a novel context-aware local region structural contrastive knowledge distillation framework (CoreKD) to enhance the distillation process for object detection. As illustrated in Fig. 1(c), with the introduction of CoreKD, the student detector can generate heatmaps that are highly consistent with those of the teacher model, leading to more accurate predictions. Our key idea is to achieve knowledge transfer at a higher level of granularity (e.g., patches), which not only mitigates the influence of background noise but also enhances the model's ability to capture local semantic representations. To this end, our framework introduces a novel patch-based semantic structural distillation (PSD) method for local knowledge transfer. This method integrates both semantic and structural information into a contrastive learning process, enabling the student model to learn stronger discriminative capabilities from both positive and negative patches. Furthermore, to explicitly model region-level contextual relationships, we propose a region-wise contextual constraint (RCC) mechanism, which enhances the student model's ability to capture global semantic dependencies. Specifically, the RCC mechanism incorporates intra-region contextual (Ita-RC) and inter-region contextual (Ite-RC) constraints into the PSD, enabling the student model to effectively model global contextual dependencies. To validate the efficacy of CoreKD, we have conducted extensive experiments on the MS COCO 2017 [31] and PASCAL VOC datasets [32]. We conduct comparative experiments across various detectors, backbone networks, and parameter settings, and the results consistently confirm the effectiveness of our proposed CoreKD framework. Specifically, on the MS COCO 2017 dataset, applying CoreKD results in mean average precisions (mAPs) of 43.6%, 40.8%, 39.8%, and 42.4% for GFL [7], Faster R-CNN [5], RetinaNet [6], and FCOS [8] detectors with ResNet-50 [9] backbones, respectively. These results surpass other state-of-the-art KD methods and show improvements over the original detectors by 3.4%, 2.4%, 2.4%, and 3.9%, respectively. In summary, the main contributions of this article are as follows.

1) We propose a novel KD framework, CoreKD, for object detection that effectively transfers the teacher detector's feature representation ability to the student detector.
2) We introduce a local region semantic structural contrastive distillation (PSD) method that guides the student model in learning localized discrimination knowledge from the teacher model. We also integrate intra-region and Ite-RC constraints to enhance the student detector's capacity to model global contextual dependencies.
3) Experiments on public MS COCO 2017 and PASCAL VOC datasets demonstrate that our proposed CoreKD outperforms the state-of-the-art KD methods on object detection. For example, CoreKD enhances the performance of the GFL detector with a ResNet-50 backbone on MS COCO 2017, achieving a mAP of 43.6%, which represents a 3.6% improvement over the original detector.

The rest of this article is organized as follows. Section II provides a brief review of related work, including object detection, KD, and contrastive learning. Section III details the proposed CoreKD framework, including the local region semantic structural contrastive KD method, the Ita-RC constraint, the Ite-RC constraint, and the overall training loss functions. Section IV covers the experimental setup, results, and associated analyses. Finally, Section V presents the conclusions.

II. Related Works

A. Object Detection

As a fundamental visual task, object detection has applications across numerous fields. The rapid advancements in deep learning have significantly propelled the progress of object detection. The convolutional neural networks (CNNs), known for their powerful local feature modeling capabilities, are particularly favored for object detection tasks. The CNN-based detectors can be categorized into single-stage and two-stage models based on whether they utilize a region proposal network (RPN) [5] for the region of interest selection. Due to the RPN's ability to distinguish between foreground and background candidates, two-stage models [5], [34], [35] typically exhibit superior detection performance. However, introducing the RPN also limits the inference speed of two-stage models. For example, the popular Faster R-CNN detector [5] first

TABLE I
COMPARISON OF DIFFERENT DISTILLATION PARADIGMS IN OBJECT DETECTION

Types	Core Mechanism	Contrastive	Param-Free
Pixel-wise [10], [17]	L_2 Distance	×	✓
Region-based [20], [22], [24]	Masking / Attention	×	✓
Generative [29]	Diffusion Process	×	×
Patch Align. [42], [45]	Spatial Alignment	×	×
CoreKD	Relation & Contrast	✓	✓

used an RPN network to simultaneously predict object bounds and objectness scores at each position, generating high-quality region proposals. These proposals are then refined using a Fast R-CNN detector. Although single-stage models [7], [36], [37], [38] offer faster inference speeds, they face the challenge of selecting positive and negative samples from dense predictions. To address this challenge, strategies for positive and negative sample selection have been developed, such as adaptive training sample selection (ATSS) [36], probabilistic anchor assignment [37], and task alignment learning [38]. In addition, to overcome the issue of additional hyperparameters in anchor design, anchor-free methods have been introduced to further improve these detectors.

These meticulously designed detectors exhibit highly satisfactory performance in object detection. Although specific designs, such as anchor-free methods, attempt to reduce the computational parameters of existing detectors, the bulk of the parameters still originate from the heavy backbone networks used to ensure robust feature representation. Therefore, optimizing the parameters of detectors while maintaining detection performance has become a crucial research direction recently.

B. Knowledge Distillation

The advancement of deep learning has led to remarkable improvements in model performance, accompanied by a continuous increase in network depth. For instance, in object detection, the earliest detectors used a 16-layer VGG [39] as the backbone, whereas current detectors employ 101-layer ResNet [9] or ResNeXt [40] backbones. KD [10] aims to transfer knowledge from a well-trained, heavy, and high-performance model (teacher) to a lightweight model (student), thereby achieving model compression. In object detection, existing KD methods include feature distillation and logits distillation [30]. Feature distillation primarily involves minimizing the distance between the teacher model and the student model in the feature space. Due to the sensitivity of object detection to positional information, current feature distillation methods focus on the selection of distillation regions, such as instance-level regions [18], foreground/background [20], and predicted regions [24]. Recently, utilizing structured information [19], [23] for KD has demonstrated impressive performance. However, these methods operate across the entire feature space in a pixel-pixel manner, which may not be conducive to understanding the global context due to the presence of substantial background information. Besides, pixel-level distillation also ignores the semantic information of the neighboring pixels. To overcome these limitations, several works have explored patch-based KD, primarily in classification [41], segmentation [42], and person reidentification [43] tasks. For example, PatKD [43] proposed to distill patch-level identity features for lifelong reidentification, while TransKD [42] applied patch-wise token alignment in transformer-based segmentation. Some works also explore intrasample and intersample relational constraints [44], or design cross-layer alignment mechanisms for patch representation transfer [45]. However, these methods are often tailored for tasks with dense supervision or regular spatial semantics, and typically assume fixed spatial correspondences between teacher and student models. To explicitly distinguish CoreKD from these existing paradigms, particularly regarding the underlying mechanism and computational efficiency, we summarize the properties of representative methods in Table I.

In contrast, CoreKD addresses these challenges with a task-specific, lightweight, and unified distillation framework. It introduces two complementary components: a PSD module that enhances structural discriminability through contrastive learning, and a RCC module that models both intra- and inter-region dependencies. Unlike previous works, which rely on additional relation modules or cross-layer alignment, CoreKD is plug-and-play and does not require modifications to the model architecture or training pipeline. All operations are performed at the feature level using native tensor reshaping and region grouping, ensuring broad applicability to various detectors and backbones.

C. Contrastive Learning

Contrastive learning has gained significant popularity among researchers due to its superior performance in image representation tasks [46], [47]. The core principle of contrastive learning is to minimize the distance between positive sample pairs while maximizing the distance between negative sample pairs. For example, it has been used to train models under unsupervised settings by bringing features of different augmented views of the same image closer while pushing features of different images apart [47]. This enables learning of high-quality visual feature representations. In addition, contrastive learning has been applied to map images and texts into a shared embedding space, enabling the training of general vision models capable of understanding vision-language relationships in zero-shot scenarios [48]. In the field of KD, contrastive learning has been used to develop a bidirectional KD approach using online contrastive objectives, improving performance in visual recognition tasks through dynamic collaboration between teacher and student models [49].

Unlike previous methods, we propose a novel semantic structural contrastive knowledge transfer approach by integrating local semantics and structural information into the distillation. This method allows the student detector to learn consistent knowledge from positive patches and diverse knowledge from negative patches, which could enhance the discrimination capability of students.

III. METHOD

A. Overview of the CoreKD

To fully leverage context-aware localized region information, we propose a novel KD framework (as shown in Fig. 2), CoreKD, for object detection tasks. In this framework, we first adopt a PSD method that guides student detectors' learning. In addition, we incorporate a RCC to improve knowledge transfer efficacy and help the student model develop contextual dependency modeling abilities. During the training phase, our proposed distillation loss, combined with the original detection losses, constrains the student detector. In the testing phase, only the trained student detector is used for inference, achieving the goal of model compression.

B. Preliminaries

The traditional feature-based distillation is to minimize the distance between the multiscale features extracted by the teacher detector and the student detector. During this process, the l_p norm is typically adopted to measure the distance between teacher and student models. Given an input image x, let the teacher detector be $f_T(x)$, and the student detector be $f_S(x)$. Note that here, we only consider the feature extraction parts of the student and teacher detectors. We can formulate this as follows:

$$\mathcal{L}_{KD} = \sum_{i=1}^{M} \left\| f_T^i(x) - g_i\left(f_S^i(x)\right)\right\|_p \tag{1}$$

where $\mathcal{L}_{KD}$ represents the KD loss, $f_T^i(x)$ and $f_S^i(x)$ denote the features extracted at the ith layer of the teacher and student detectors, respectively, g_i is the adaptive layer applied to the student detector's features at the ith layer, M is the number of layers, and $\|\cdot\|_p$ is the l_p-norm used to measure the distance between the teacher and student features.

C. Patch-Based Semantic Structural Contrastive Distillation

In the context of object detection, pixel-level distillation may not be optimal, as it introduces interference from background pixels and overlooks valuable local structural knowledge. Instead, we propose transferring knowledge at a higher granularity level, such as patches [50], which can simultaneously mitigate background interference and model local semantic information. However, a natural question remains: *How to effectively transfer the patch-wise knowledge*. To address this, we propose a novel patch-based semantic structural contrastive distillation method that integrates local region semantics and structural information. This method enables the student detector to learn both similarities (consistency) from positive patches and differences (diversity) from negative patches.

Specifically, we first segment the intermediate layer features $F^T \in \mathbb{R}^{C\times H\times W}$ and $F^S \in \mathbb{R}^{C\times H\times W}$, from the teacher and student detectors, into nonoverlapping local feature patches $P^T \in \mathbb{R}^{N\times D}$ and $P^S \in \mathbb{R}^{N\times D}$, resulting in N patches each with D dimensions, $D = p_s \times p_s \times C$ and p_s is the patch size. We then treat corresponding feature patches from the teacher and student detectors as positive pairs (P_i^T, P_i^S), while noncorresponding feature patches are treated as negative pairs $(P_i^T, P_j^S), i \neq j$. Then, we measure the similarity between positive and negative pairs to construct the contrastive loss. We consider both semantic and structural similarities to formulate the similarity matrices for the positive and negative pairs. Specifically, we directly multiply the P^T and the transpose of P^S to construct the semantic similarity matrix $S_e \in \mathbb{R}^{N\times N}$. In addition, we compute the structural similarity index measure (SSIM) [51] between P^T and P^S to obtain the structural similarity matrix S_t. Given the computational complexity of SSIM, we calculate it only for the positive pairs and diagonalize it. The calculation process is as follows:

$$\begin{aligned} S_e &= P^T \cdot \left(P^S\right)^T \\ S_t &= diag\left(\frac{\left(2\mu_{P_i^T}\mu_{P_i^S} + C_1\right)\left(2\sigma_{P_i^T P_i^S} + C_2\right)}{\left(\mu_{P_i^T}^2 + \mu_{P_i^S}^2 + C_1\right)\left(\sigma_{P_i^T}^2 + \sigma_{P_i^S}^2 + C_2\right)}\right) \end{aligned} \tag{2}$$

where i is the index of patches, $\mu_{P_i^T}$ and $\mu_{P_i^S}$ are the means of the matrices P_i^T and P_i^S, respectively, $\sigma_{P_i^T}$ and $\sigma_{P_i^S}$ are the variances of the matrices P_i^T and P_i^S, respectively, and $\sigma_{P_i^T P_i^S}$ is the covariance of the matrices P_i^T and P_i^S. Constants C_1 and C_2 are used to stabilize the calculations, and following the original work [51], we set them to 0.0001 and 0.0009, respectively. Besides, $(\cdot)^T$ indicates the transpose of a matrix, and *diag* is the diagonalization. To consider both semantic and structural similarity, we reconstruct the similarity matrix $S \in \mathbb{R}^{N\times N}$

$$S^{i,j} = \begin{cases} \frac{1}{2}\left(S_e^{i,i} + S_t^{i,i}\right), & \text{if } i = j \\ S_e^{i,j}, & \text{if } i \neq j. \end{cases} \tag{3}$$

Furthermore, we can construct our contrastive loss [47], [48] based on the similarity matrix S. Positive knowledge patches are labeled as 1, while negative ones are labeled as 0. Thus, the calculation of our local semantic structural contrastive distillation loss is as follows:

$$\begin{aligned} \mathcal{L}_{psd} = &-\frac{1}{2N}\sum_{i=1}^{N}\log\frac{\exp\left(S^{i,i}/\tau_s\right)}{\sum_{j=1}^{N}\exp\left(S^{i,j}/\tau_s\right)} \\ &-\frac{1}{2N}\sum_{j=1}^{N}\log\frac{\exp\left(S^{j,j}/\tau_s\right)}{\sum_{i=1}^{N}\exp\left(S^{i,j}/\tau_s\right)} \end{aligned} \tag{4}$$

where σ represents the softmax function, $S^{i,j}$ is the element of the similarity matrix S at position (i, j), and τ_s is a temperature parameter. Compared to other pixel-level KD frameworks, our proposed PSD framework incorporates semantic structural information in a contrastive manner.

D. Contextual Knowledge Constraint

While the PSD framework enables effective patch-level knowledge transfer, an overemphasis on local knowledge during the distillation process may inadvertently hinder the student model's ability to capture global contextual relationships. Contextual constraints [19], [20] have been demonstrated to play a crucial role in enhancing the learning of student models. However, existing methods primarily focus on modeling pixel-level contextual relationships, such as nonlocal blocks [52] and GCBlock [53]. To address this, we introduce a region-wise contextual constraint to ensure the student model develops the ability to model contextual dependencies through the PSD framework. Specifically, we introduce both Ita-RC and Ite-RC constraints to enhance the student model's ability to model patch-wise contextual dependencies.

1) Ita-RC Constraint: To capture intra-region context, we aggregate all pixels within each region individually. We then minimize the distance between the aggregated Ita-RC knowledge of the teacher and student models to achieve knowledge transfer.

Given the local feature $P^T \in \mathbb{R}^{N\times D}$ or $P^S \in \mathbb{R}^{N\times D}$ from the teacher and student models, we first need to aggregate all patches and take their average

$$\bar{A}_i^T = \frac{1}{D}\sum_{d=1}^{D}\left|P_{i,d}^T\right|, \quad \bar{A}_i^S = \frac{1}{D}\sum_{d=1}^{D}\left|P_{i,d}^S\right| \tag{5}$$

where $\bar{A}_i^T$ and $\bar{A}_i^S$ represent the aggregated features of the ith patch for the teacher and student models, respectively. $P_{i,d}^T$ and $P_{i,d}^S$ are the dth elements of the ith local feature patch for the teacher and student models. The $|\cdot|$ is the operation of absolute value, and D is the feature dimension.

Then, we need to normalize the aggregated features $\bar{A}^T$ and $\bar{A}^S$ using the softmax function

$$\tilde{A}_{N_i}^T = \frac{\exp\left(\bar{A}_i^T\right)/\tau_s}{\sum_{j=1}^{N}\exp\left(\bar{A}_j^T\right)}, \quad \tilde{A}_{N_i}^S = \frac{\exp\left(\bar{A}_i^S\right)/\tau_s}{\sum_{j=1}^{N}\exp\left(\bar{A}_j^S\right)} \tag{6}$$

where $\tilde{A}_{N_i}^T$ and $\tilde{A}_{N_i}^S$ represent the normalized features of the teacher and student models, respectively. $\sum_{j=1}^{N}$ denotes the summation over all N patches.

We compute the mean squared error (mse) loss between the normalized Ita-RC features of the teacher and student models

$$\mathcal{L}_{ita} = \frac{1}{N}\sum_{i=1}^{N}\left|\tilde{A}_{N_i}^T - \tilde{A}_{N_i}^S\right|^2. \tag{7}$$

2) Ite-RC Constraint: Based on the Ita-RC, it is straightforward to extend this approach to Ite-RC constraints. Similarly, we aggregate features of different patches at the patch level to capture inter-region context. And then we can constrain the Ite-RC knowledge between the teacher and student models.

Given the local feature patches $P_i^T \in \mathbb{R}^{1\times D}$ or $P_i^S \in \mathbb{R}^{1\times D}$ from the teacher and student models, we first aggregate all the patches and take their average

$$\bar{E}^T = \frac{1}{N}\sum_{i}^{N}\left|P_i^T\right|, \quad \bar{E}^S = \frac{1}{N}\sum_{i}^{N}\left|P_i^S\right| \tag{8}$$

where $\bar{E}^T$ and $\bar{E}^S$ represent the aggregated inter-region context features of the teacher and student models, respectively. N is the total number of feature patches.

Next, we normalize the aggregated features using the softmax function

$$\tilde{E}_{N_i}^T = \frac{\exp\left(\bar{E}_{N_i}^T\right)/\tau_s}{\sum_{d=1}^{D}\exp\left(\bar{E}_{N_d}^T\right)}, \quad \tilde{E}_{N_i}^S = \frac{\exp\left(\bar{E}_{N_i}^S\right)/\tau_s}{\sum_{d=1}^{D}\exp\left(\bar{E}_{N_d}^S\right)} \tag{9}$$

where $\tilde{E}_{N_i}^T$ and $\tilde{E}_{N_i}^S$ are the normalized inter-region context features of the teacher and student models, respectively. $\bar{E}_{N_i}^T$ and $\bar{E}_{N_i}^S$ are the elements at the ith position of the aggregated features, $\sum_{d=1}^{D}$ denotes the summation over all feature dimensions D, and τ_s is a temperature parameter.

Finally, we compute the mse loss between the normalized Ite-RC features of the teacher and student models

$$\mathcal{L}_{ite} = \frac{1}{D}\sum_{i=1}^{D}\left|\tilde{E}_{N_i}^T - \tilde{E}_{N_i}^S\right|^2. \tag{10}$$

3) Region-Wise Contextual Knowledge Constraint: We combine the Ite-RC and Ita-RC constraints to form our final region-wise contextual constraint, which is computed as follows:

$$\mathcal{L}_{rcc} = \mathcal{L}_{ite} + \mathcal{L}_{ita}. \tag{11}$$

E. Overall Loss Function

To summarize, we incorporate our proposed PSD loss and RCC loss as the distillation losses and jointly train the detector with the original detection losses (including localization and classification losses). The overall loss function is computed as follows:

$$\mathcal{L}_{all} = \mathcal{L}_{det} + \lambda_1\mathcal{L}_{psd} + \lambda_2\mathcal{L}_{rcc} \tag{12}$$

where $\mathcal{L}_{det}$ represents the original detection loss, λ_1 and λ_2 are hyper-parameters used to adjust the contributions of the $\mathcal{L}_{psd}$ and $\mathcal{L}_{rcc}$ losses, respectively.

IV. Experiments

A. Experimental Setup

1) MS COCO: We conduct experimental validation of our proposed framework on the widely used MS COCO dataset [31]. Consistent with previous work [20], we use the 120-K training images to train the student detector and the 5-K validation images to evaluate the model's performance. To provide a comprehensive assessment, we employ six metrics: mAP to evaluate overall performance, AP_{50} and AP_{75} to measure precision at different IoU thresholds, and AP_S, AP_M, and AP_L to assess performance across small, medium, and large objects, respectively.

2) Pascal VOC: We further validate the effectiveness of the CoreKD framework using the PASCAL VOC dataset [32]. The VOC 07 + 12 training protocol is employed, which combines the VOC 2007 trainval set and the VOC 2012 trainval set (totaling 16 551 images) for training, with evaluation performed on the VOC 2007 test set (4952 images). To thoroughly assess localization performance, we report the average precision (AP) at different IoU thresholds, specifically AP_{50}, AP_{60}, AP_{70}, AP_{80}, and AP_{90}.

3) VisDrone: To further evaluate the generalizability of the CoreKD framework, we conduct experiments on the VisDrone dataset [61], a challenging benchmark featuring aerial imagery captured by drones in urban environments. The dataset contains a wide range of object scales, dense object distributions, and significant occlusions. Following standard practice, we use the VisDrone2019-DET train set (6471 images) for training and report results on the VisDrone2019-DET validation set (548 images). We evaluate performance using AP metrics at IoU thresholds of 0.5 and 0.75, along with the overall AP.

TABLE II

COMPARISON OF COREKD WITH OTHER KD METHODS ON THE MS COCO DATASET USING THE GFL DETECTOR. BLACK HIGHLIGHTED FONTS REPRESENT THE HIGHEST VALUES. THE VALUES HIGHLIGHTED IN BLUE ARE THOSE THAT HAVE INCREASED THE MOST. * DENOTES OUR REPRODUCED RESULTS

Method	Backbones	Publications	Years	Schedule	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
GFL [7] (T)	ResNet-101	NeurIPS	2020	24	44.9	63.1	49.0	28.0	49.1	57.2
GFL [7] (S)	ResNet-50	NeurIPS	2020	12	40.2	58.4	43.3	23.3	44.0	52.2
FitNets [10]	ResNet-50	NeurIPS	2015	12	40.7 (+0.5)	58.6 (+0.2)	44.0 (+0.7)	23.7 (+0.4)	44.4 (+0.4)	53.2 (+1.0)
Inside GT Box	ResNet-50	-	-	12	40.7 (+0.5)	58.6 (+0.2)	44.2 (+0.9)	23.1 (-0.2)	44.5 (+0.5)	53.5 (+1.3)
Defeat [54]	ResNet-50	CVPR	2021	12	40.8 (+0.6)	58.6 (+0.2)	44.2 (+0.9)	24.3 (+1.0)	44.6 (+0.6)	53.7 (+1.5)
Main Region [25]	ResNet-50	IEEE Tpami	2023	12	41.1 (+0.9)	58.7 (+0.3)	44.4 (+1.1)	24.1 (+0.8)	44.6 (+0.6)	53.6 (+1.4)
FGFI [55]	ResNet-50	CVPR	2019	12	41.1 (+0.9)	58.8 (+0.4)	44.8 (+1.5)	23.3 (+0.0)	45.4 (+1.4)	53.1 (+0.9)
GID [56]	ResNet-50	CVPR	2021	12	41.5 (+1.3)	59.6 (+1.2)	45.2 (+1.9)	24.3 (+1.0)	45.7 (+1.7)	53.6 (+1.4)
CMD [14]	ResNet-50	IEEE TMM	2024	12	42.0 (+1.8)	59.7 (+1.3)	45.7 (+2.4)	24.2 (+0.9)	46.3 (+2.3)	54.4 (+2.2)
MGD* [22]	ResNet-50	ECCV	2022	12	42.1 (+1.9)	60.3 (+1.9)	45.8 (+2.5)	24.4 (+1.1)	46.2 (+2.2)	54.7 (+2.5)
SKD [23]	ResNet-50	NeurIPS	2022	12	42.3 (+2.1)	60.2 (+1.8)	45.9 (+2.6)	24.4 (+1.1)	46.7 (+2.7)	55.6 (+3.4)
ScaleKD [21]	ResNet-50	CVPR	2023	12	42.5 (+2.3)	-	-	25.9 (+2.6)	46.2 (+2.2)	54.6 (+2.4)
PKD* [17]	ResNet-50	NeurIPS	2022	12	42.5 (+2.3)	60.9 (+2.5)	46.0 (+2.7)	24.2 (+0.9)	46.7 (+2.7)	55.9 (+3.7)
VKD* [57]	ResNet-50	CVPR	2024	12	42.6 (+2.4)	60.7 (+2.3)	46.0 (+2.7)	24.9 (+1.6)	46.5 (+2.5)	55.8 (+3.6)
DrKD [58]	ResNet-50	PR	2025	12	42.6 (+2.4)	60.6 (+2.2)	46.1 (+2.8)	24.6 (+1.3)	47.0 (+3.0)	55.3 (+3.1)
LD [25]	ResNet-50	IEEE Tpami	2023	12	43.0 (+2.8)	61.6 (+3.2)	46.6 (+3.3)	25.5 (+2.2)	47.0 (+3.0)	55.8 (+3.6)
BCKD [28]	ResNet-50	ICCV	2023	12	43.2 (+3.0)	61.6 (+3.2)	46.9 (+3.6)	25.7 (+2.4)	47.3 (+3.3)	55.9 (+3.7)
FreeKD* [59]	ResNet-50	CVPR	2024	12	43.3 (+3.1)	61.5 (+3.1)	46.9 (+3.6)	25.4 (+2.1)	47.4 (+3.4)	**56.4** (**+4.2**)
FGD* [20]	ResNet-50	CVPR	2022	12	43.4 (+3.2)	61.7 (+3.3)	47.0 (+3.7)	26.2 (+2.9)	47.4 (+3.4)	**56.4** (**+4.2**)
CrossKD [26]	ResNet-50	CVPR	2024	12	**43.7** (**+3.5**)	**62.1** (**+3.7**)	**47.4** (**+4.1**)	**26.9** (**+3.6**)	**48.0** (**+4.0**)	56.2 (+4.0)
CoreKD (Ours)	ResNet-50	-	-	-	43.6 (+3.4)	**62.1** (**+3.7**)	47.3 (+4.0)	26.6 (+3.3)	**48.0** (**+4.0**)	**56.4** (**+4.2**)
GFLV2 [60] (T)	ResNet-101	CVPR	2021	24	45.8	63.7	50.0	27.8	50.2	58.6
GFLV2 [60] (S)	ResNet-50	CVPR	2021	12	41.0	58.5	45.0	24.3	44.6	53.4
CoreKD (Ours)	ResNet-50	-	-	-	44.3 (+3.3)	62.1 (+3.6)	47.9 (+2.9)	26.4 (+2.1)	47.8 (+3.2)	58.4 (+5.0)

4) Implementation Details: To validate the effectiveness of the proposed CoreKD framework, we select GFL [7] as our detector. In addition to incorporating the distillation losses we introduce, we adopt all network configurations and details from GFL [7]. Our experiments are conducted on a server equipped with dual NVIDIA 3090 GPUs, each with 24 GB of memory. The implementation relies on the MMDetection [62] and PyTorch [63] frameworks. We follow the training and testing strategies of the GFL detector as outlined in MMDetection, with a few modifications. Specifically, we set the batch size to 8 and use the stochastic gradient descent (SGD) [64] optimizer with an initial learning rate of 0.01 and a decay rate of 0.1 during the training process. And the pretrained checkpoint of the teacher is also provided by the MMDetection model zoo [62].

In our CoreKD framework, we configure the patch size to 7, ensuring a fine-grained analysis of local features. The temperature coefficient τ_s is set to 0.5 to adjust the distribution of the softmax function during the distillation process. We also set the regularization parameters λ_1 and λ_2 to 0.0001 and 0.001, respectively, to balance the different loss functions.

B. Comparison With the State-of-the-Art

The experimental results presented in Table II highlight the superiority of our proposed CoreKD framework in comparison to 16 other KD methods when applied to the GFLV1 [7] detector. In our setup, the teacher detector utilizes a GFL detector with a ResNet-101 backbone, while the student detector employs a GFL detector with a ResNet-50 backbone. Besides, the training schedule is 12 epochs.

As shown in Table II, our proposed CoreKD attains an AP_{50} of 62.1, an AP_{75} of 47.3, an AP_S of 26.6, an AP_M of 48.0, and an AP_L of 56.4. Compared with other KD methods, CoreKD achieves the best performance in six reported metrics. These results highlight the robustness of CoreKD across IoU thresholds and object sizes, demonstrating its effectiveness in object detection tasks. More specifically, CoreKD achieves a mAP of 43.6, which is a significant improvement over the student model with a mAP of 40.2 (+3.4). When compared to other state-of-the-art KD methods, CoreKD also demonstrates superior performance: FitNets (+2.9) [10], Inside GT Box (+2.9), Defeat (+2.8) [54], Main Region (+2.5) [25], FGFI (+2.5) [55], GID (+2.1) [56], CMD (+1.4) [14], MGD (+1.5) [22], SKD (+1.3) [23], ScaleKD (+1.1) [21], PKD (+1.1) [17], VKD (+1.0) [57], DrKD (+1.0) [58], LD (+0.6) [25], BCKD (+0.4) [28], FreeKD (+0.3) [59], and FGD (+0.2) [20]. We also note that CrossKD and CoreKD achieve comparable performance on GFL (43.7 versus 43.6 mAP), but CoreKD exhibits stronger generalization across different detection frameworks. Especially, CoreKD surpasses CrossKD by +1.1 mAP on FCOS (see in Table III), while requiring no architectural modifications or auxiliary prediction heads. This suggests that CoreKD provides a more general and lightweight solution, particularly suitable for diverse object detectors in real-world applications.

For completeness, we further validate CoreKD on the improved GFLv2 framework [60]. As shown in Table II, CoreKD consistently enhances the performance of the student, yielding improvements of +3.3 mAP, +3.6 AP_{50}, +2.1 AP_S, +3.2 AP_M, and +5.0 AP_L compared to the baseline. These results demonstrate that CoreKD is not limited to a specific detector version and remains effective under a stronger baseline.

These results demonstrate that our proposed CoreKD effectively facilitates knowledge transfer and significantly enhances the performance of student models. Moreover, these results

TABLE III

COMPARISON OF COREKD WITH OTHER KD METHODS ON THE MS COCO DATASET ACROSS VARIOUS DETECTOR FRAMEWORKS. THE VALUES HIGHLIGHTED IN BLUE REPRESENT THE LARGEST IMPROVEMENTS OVER THE BASELINE STUDENT MODEL

Method	Backbones	Publications	Years	Schedule	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Two-stage detectors										
Faster RCNN [5] (T)	ResNet-101	NeurIPS	2015	24	39.8	60.1	43.3	22.5	43.6	52.8
Faster RCNN [5] (S)	ResNet-50	NeurIPS	2015	24	38.4	59.0	42.0	21.5	42.1	50.3
FitNet [10]	ResNet-50	NeurIPS	2015	24	38.9 (+0.5)	59.5 (+0.5)	42.4 (+0.4)	21.9 (+0.4)	42.2 (+0.1)	51.6 (+1.3)
FGFI [55]	ResNet-50	CVPR	2019	24	39.3 (+0.9)	59.8 (+0.8)	42.9 (+0.9)	22.5 (+1.0)	42.3 (+0.2)	52.2 (+1.9)
FRS [65]	ResNet-50	NeurIPS	2021	24	39.5 (+1.1)	60.1 (+1.1)	43.3 (+1.3)	22.3 (+0.8)	43.6 (+1.5)	51.7 (+1.4)
FTPD [66]	ResNet-50	TCSVT	2025	24	39.7 (+1.3)	60.1 (+1.1)	43.2 (+1.2)	-	-	-
CKD [67]	ResNet-50	TIP	2025	24	39.7 (+1.3)	60.6 (+1.6)	43.3 (+1.3)	-	-	-
FBKD [68]	ResNet-50	ICLR	2020	24	40.2 (+1.8)	60.4 (+1.4)	43.6 (+1.6)	22.8 (+1.3)	43.8 (+1.7)	53.2 (+2.9)
GID [56]	ResNet-50	CVPR	2021	24	40.2 (+1.8)	60.8 (+1.8)	43.6 (+1.6)	**23.6 (+2.1)**	43.9 (+1.8)	53.0 (+2.7)
DeFeat [54]	ResNet-50	CVPR	2021	24	40.3 (+1.9)	60.9 (+1.9)	44.0 (+2.0)	23.1 (+1.6)	44.1 (+2.0)	53.4 (+3.1)
FGD [20]	ResNet-50	CVPR	2022	24	40.5 (+2.1)	-	-	22.6 (+1.1)	44.7 (+2.6)	53.2 (+2.9)
GKD [69]	ResNet-50	ECCV	2022	24	40.6 (+2.2)	61.0 (+2.0)	44.0 (+2.0)	23.4 (+1.9)	44.4 (+2.3)	53.3 (+3.0)
DiffKD [29]	ResNet-50	NeurIPS	2023	24	40.6 (+2.2)	60.9 (+1.9)	43.9 (+1.9)	23.0 (+1.5)	44.5 (+2.4)	54.0 (+3.7)
DCSF-KD [70]	ResNet-50	AAAI	2025	24	40.7 (+2.3)	61.0 (+2.0)	**44.6 (+2.6)**	23.2 (+1.7)	45.0 (+2.9)	53.5 (+3.2)
CoreKD (Ours)	ResNet-50	-	-	24	**40.8 (+2.4)**	**61.1 (+2.1)**	**44.6 (+2.6)**	23.0 (+1.5)	**45.1 (+3.0)**	**54.5 (+4.2)**
One-stage detectors										
RetinaNet [6] (T)	ResNet-101	ICCV	2017	24	38.9	58.0	41.5	21.0	42.8	52.4
RetinaNet [6] (S)	ResNet-50	ICCV	2017	24	37.4	56.7	39.6	20.6	40.7	49.7
FitNet [10]	ResNet-50	NeurIPS	2015	24	37.4 (+0.0)	57.1 (+0.4)	40.0 (+0.4)	20.8 (+0.2)	40.8 (+0.1)	50.9 (+1.2)
FGFI [55]	ResNet-50	CVPR	2019	24	38.6 (+1.2)	58.7 (+2.0)	41.3 (+1.7)	21.4 (+0.8)	42.5 (+1.8)	51.5 (+1.8)
DenseKD [71]	ResNet-50	IEEE TNNLS	2025	24	38.6 (+1.2)	58.8 (+2.1)	41.7 (+2.1)	**22.8 (+2.2)**	43.1 (+2.4)	50.1 (+0.4)
GID [56]	ResNet-50	CVPR	2021	24	39.1 (+1.7)	59.0 (+2.3)	42.3 (+2.7)	**22.8 (+2.2)**	43.1 (+2.4)	52.3 (+2.6)
DeFeat [54]	ResNet-50	CVPR	2021	24	39.3 (+1.9)	58.2 (+1.5)	42.1 (+2.5)	21.7 (+1.1)	42.9 (+2.2)	52.9 (+3.2)
FBKD [68]	ResNet-50	ICLR	2020	24	39.3 (+1.9)	58.8 (+2.1)	42.0 (+2.4)	21.2 (+0.6)	43.2 (+2.5)	53.0 (+3.3)
FRS [65]	ResNet-50	NeurIPS	2021	24	39.3 (+1.9)	58.8 (+2.1)	42.0 (+2.4)	21.5 (+0.9)	43.3 (+2.6)	52.6 (+2.9)
PFI [72]	ResNet-50	AAAI	2022	24	39.6 (+2.2)	-	-	21.4 (+0.8)	**44.0 (+3.3)**	52.5 (+2.8)
CrossKD [26]	ResNet-50	CVPR	2024	24	39.7 (+2.3)	**58.9 (+2.2)**	42.5 (+2.9)	22.4 (+1.8)	43.6 (+2.9)	52.8 (+3.1)
FGD [20]	ResNet-50	CVPR	2022	24	39.7 (+2.3)	-	-	22.0 (+1.4)	43.7 (+3.0)	**53.6 (+3.9)**
DiffKD [29]	ResNet-50	NeurIPS	2023	24	39.7 (+2.3)	58.6 (+1.9)	42.1 (+2.5)	21.6 (+1.0)	43.8 (+3.1)	53.3 (+3.6)
CoreKD (Ours)	ResNet-50	-	-	24	**39.8 (+2.4)**	**58.9 (+2.2)**	**42.6 (+3.0)**	22.0 (+1.4)	43.8 (+3.1)	53.5 (+3.8)
Anchor-free detectors										
FCOS [8] (T)	ResNet-101	IEEE Tpami	2020	24	40.8	60.0	44.0	24.2	44.3	52.4
FCOS [8] (S)	ResNet-50	IEEE Tpami	2020	24	38.5	57.7	41.0	21.9	42.8	48.6
FRS [65]	ResNet-50	NeurIPS	2021	24	40.9 (+2.4)	60.3 (+2.6)	43.6 (+2.6)	25.7 (+3.8)	45.2 (+2.4)	51.2 (+2.6)
CrossKD [26]	ResNet-50	CVPR	2024	24	41.3 (+2.8)	60.6 (+2.9)	44.2 (+3.2)	25.1 (+3.2)	45.5 (+2.7)	52.4 (+3.8)
FGD [20]	ResNet-50	CVPR	2022	24	42.1 (+3.6)	-	-	**27.0 (+5.1)**	**46.0 (+3.2)**	54.6 (+6.0)
DiffKD [29]	ResNet-50	NeurIPS	2023	24	**42.4 (+3.9)**	**61.0 (+3.3)**	**45.8 (+4.8)**	26.6 (+4.7)	45.9 (+3.1)	54.8 (+6.2)
CoreKD (Ours)	ResNet-50	-	-	24	**42.4 (+3.9)**	**61.0 (+3.3)**	45.7 (+4.7)	26.9 (+5.0)	**46.0 (+3.2)**	**55.0 (+6.4)**

confirm that the proposed local region contrastive distillation and contextual knowledge constraints effectively focus on local semantic knowledge and contextual information, thereby improving the KD process in object detection tasks.

C. Comparative Experiments on Other Detectors

We also compare the performance of CoreKD with other state-of-the-art KD methods across various detector frameworks (e.g., two-stage, one-stage, and anchor-free), as shown in Table III. We use ResNet-101 as the backbone for the teacher detector and ResNet-50 for the student detector. The student detector is trained with a 24-epoch schedule.

1) Two-Stage Detectors: CoreKD demonstrates superior performance across multiple metrics in Faster R-CNN detector (two-stage), including mAP, AP_{50}, AP_{75}, AP_M, and AP_L, as shown in Table III. Specifically, CoreKD achieves a mAP of 40.8, representing a significant improvement over the baseline student model by 2.4 points. In addition, CoreKD also improves AP_{50} by 2.1 points and AP_{75} by 2.6 points. When compared to other state-of-the-art KD methods, as shown in Table III, CoreKD outperforms FitNet [10] by 1.9 points (40.8 versus 38.9), FGFI [55] by 1.5 points (40.8 versus 39.3), FRS [65] by 1.3 points (40.8 versus 39.5), FTPD [66] by 1.1 points (40.8 versus 39.7), CKD [67] by 1.1 points (40.8 versus 39.7), FBKD [68] by 0.6 points (40.8 versus 40.2), GID [56] by 0.6 points (40.8 versus 40.2), DeFeat [54] by 0.5 points (40.8 versus 40.3), FGD [20] by 0.3 points (40.8 versus 40.5), GKD [69] by 0.2 points (40.8 versus 40.6), DiffKD [29] by 0.2 points (40.8 versus 40.6), and DCSF-KD [70] by 0.1 points (40.8 versus 40.7).

2) One-Stage Detectors: In the context of one-stage detector RetinaNet, CoreKD achieves a mAP of 39.8, significantly outperforming the baseline student model by 2.4 points, as shown in Table III. Beyond mAP, CoreKD exhibits competitive or superior performance in other critical metrics, improving AP_{50} by 2.2 points (58.9 versus 56.7) and AP_{75} by 3.0 points (42.6 versus 39.6). Furthermore, CoreKD shows marked improvements over several other state-of-the-art KD methods, as shown in Table III: it exceeds FitNet [10] by 2.4 points (39.8 versus 37.4), FGFI [55] by 1.2 points (39.8 versus 38.6), DenseKD [71] by 1.2 points (39.8 versus 38.6), GID [56] by 0.7 points (39.8 versus 39.1), DeFeat [54] by 0.7 points

TABLE IV

COMPARISON OF AP, AP_{50}, AND AP_{75} ON THE PASCAL VOC AND VISDRONE DATASETS USING GFL AND FASTER R-CNN DETECTORS. THE TEACHER IS RESNET-50 AND THE STUDENT IS RESNET-18

Model	GFL			Faster R-CNN		
	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
VisDrone						
Teacher	17.3	30.2	17.8	17.8	31.1	18.3
Student	14.7	25.9	15.0	15.7	28.2	15.9
CoreKD	15.8 **(+1.1)**	27.5	16.3	16.3 **(+0.6)**	29.0	16.5
PASCAL VOC						
Teacher	58.3	77.0	58.9	56.5	78.4	57.7
Student	54.5	74.2	55.2	52.8	76.3	52.8
CoreKD	59.6 **(+5.1)**	77.6	60.1	56.0 **(+3.2)**	77.9	56.3

TABLE V

ABLATION STUDIES ON THE MS COCO DATASET. THIS TABLE REPORTS THE EFFECTS OF DIFFERENT COMPONENTS ON THE PERFORMANCE OF THE STUDENT DETECTOR GFL

$\mathcal{L}_{psd}$	$\mathcal{L}_{ita}$	$\mathcal{L}_{ite}$	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
			40.2	58.4	43.3	23.3	44.0	52.2
✓			42.0	60.2	45.5	24.8	45.9	54.4
✓	✓		42.5	60.7	46.0	24.8	46.6	55.6
✓		✓	43.4	61.7	47.0	25.1	47.4	**57.1**
✓	✓	✓	**43.6**	**62.3**	**47.8**	**26.6**	**48.2**	56.9

(39.8 versus 39.1), FBKD [68] by 0.5 points (39.8 versus 39.3), FRS [65] by 0.5 points (39.8 versus 39.3), PFI [72] by 0.2 points (39.8 versus 39.6), CrossKD [26] by 0.1 points (39.8 versus 39.7), FGD [20] by 0.1 points (39.8 versus 39.7), and DiffKD [29] by 0.1 points (39.8 versus 39.7).

3) Anchor-Free Detectors: For anchor-free detectors such as FCOS, as shown in Table III, CoreKD achieves an impressive mAP of 42.4, matching the highest-performing method, DiffKD [29], and significantly surpassing the baseline student model by 3.9 points. This notable improvement underscores the effectiveness of CoreKD in handling anchor-free detection tasks. Moreover, CoreKD demonstrates considerable advantages over several other prominent KD methods, as shown in Table III. It surpasses FRS [65] by 1.5 points (42.4 versus 40.9), CrossKD [26] by 1.1 points (42.4 versus 41.3), and FGD [20] by 0.3 points (42.4 versus 42.1). Beyond mAP, CoreKD also excels in additional performance metrics, showcasing significant enhancements in AP_{50}, AP_{75}, and other key indicators. It is also worth highlighting the efficiency advantage of CoreKD over recent generative distillation methods, such as DiffKD [29]. Generative approaches typically introduce significant computational overhead due to the requirement of auxiliary autoencoders and iterative diffusion steps during training. In contrast, CoreKD is parameter-free and implemented via native tensor operations, avoiding the heavy computational burden of generative models. This makes CoreKD a more practical solution for real-world applications where training resources are constrained.

4) Performance on Pascal VOC Dataset Across Detectors: The experimental results presented in Table IV indicate the effectiveness of the CoreKD method in enhancing student model performance across different detectors on the PASCAL VOC Dataset. For the GFL detector, CoreKD improves the student's AP from 54.5 to 59.6, AP_{50} from 74.2 to 77.6, and AP_{75} from 55.2 to 60.1, surpassing even the teacher model's performance. This represents an increase of 5.1 in AP, 3.4 in AP_{50}, and 4.9 in AP_{75} compared to the student model without CoreKD. Similarly, for the Faster R-CNN detector, CoreKD boosts the student's AP from 52.8 to 56.0, AP_{50} from 76.3 to 77.9, and AP_{75} from 52.8 to 56.3, significantly narrowing the performance gap between the student and teacher models. Interestingly, we also noticed that in some experiments on the PASCAL VOC dataset, the student model slightly outperforms the teacher. This phenomenon might be attributed to two factors. First, VOC is a relatively small-scale dataset, which may limit the expressive capacity of large teacher models and prevent them from fully demonstrating their advantages. Second, the student not only learns from the teacher's guidance but also benefits from ground-truth supervision and the structural and contextual regularization introduced by CoreKD. These combined factors allow the student to achieve better generalization in this scenario.

5) Performance on VisDrone Dataset Across Detectors: We further evaluate CoreKD on the challenging VisDrone dataset to assess its generalizability under complex scenarios. As shown in Table IV, CoreKD consistently improves student performance across both detectors. Specifically, CoreKD improves the AP of the student model from 14.7 to 15.8 (+1.1 AP) with GFL, and from 15.7 to 16.3 (+0.6 AP) with Faster R-CNN. It is worth noting that while these absolute gains appear modest compared to other datasets, they represent a significant recovery of the teacher–student performance gap. Due to the extreme difficulty of VisDrone (e.g., tiny object scales), the teacher model (ResNet-50) achieves a relatively low upper bound (e.g., 17.3 AP for GFL), leaving a narrow distillable gap of only 2.6 AP between the teacher and the student. Consequently, the +1.1 AP improvement achieved by CoreKD corresponds to a recovery of 42.3% of the total available gap. This indicates that CoreKD is highly effective at extracting structural knowledge and narrowing the performance gap, even when the teacher provides weak or noisy supervision.

These results highlight the adaptability of CoreKD in different types of frameworks, establishing it as a competitive and effective option among current KD methods. The successful integration of PSD and RCC significantly enhances the student detector's learning capability.

D. Ablation Studies

To validate the effectiveness of each component in our proposed CoreKD framework, we conduct a series of ablation studies, as shown in Table V. We chose the GFL detector as our base student model to complete this. The first row represents the performance of the GFL without any KD. Table V summarizes the performance impact of removing individual components.

1) Impact of PSD: Introducing the $\mathcal{L}_{psd}$ loss results in a mAP of 42.0, an increase of 1.8 points over the student detector, as shown in Table V. This setup also improves AP_{50} by 1.8 points (60.2 versus 58.4) and AP_{75} by 2.2 points (45.5 versus 43.3), demonstrating the fundamental benefit of the local region semantic structural contrastive distillation framework in enhancing the learning capability of the student detector.

2) Impact of Ita-RC: Incorporating the Ita-RC knowledge constraint into the PSD framework further boosts the mAP to 42.5 (versus 42.0 with only $\mathcal{L}_{psd}$), with AP_{50} reaching 60.7 and AP_{75} achieving 46.0. These represent improvements of 2.3, 2.3, and 2.7 points over the baseline student detector (40.2, 58.4, and 43.3, respectively), as shown in Table V. This enhancement demonstrates that the Ita-RC constraint significantly improves the student's ability to understand and utilize the context within each local region, leading to better feature representation.

3) Impact of Ite-RC: Integrating the Ite-RC knowledge constraint with the PSD framework significantly enhances the performance of the student detector, as shown in Table V. This combination achieves a mAP of 43.4, an improvement of 1.4 points over using $\mathcal{L}_{psd}$ alone (42.0). In addition, AP_{50} increases to 61.7, showing a gain of 1.5 points over $\mathcal{L}_{psd}$ alone (60.2), and AP_{75} improves to 47.0, representing a 1.5-point increase (45.5 versus 47.0). These results suggest that understanding the relationships between different regions significantly contributes to the student's learning.

4) Synergistic Impact of Ita-RC and Ite-RC: We also reveal a synergy between Ita-RC and Ite-RC. The simultaneous integration of both components elevates the student model's performance to state-of-the-art levels on all evaluated metrics. Specifically, mAP, AP_{50}, AP_{75}, AP_S, and AP_M all reached new highs. The improvement is particularly pronounced for small object detection, with AP_S increasing by 1.5 (26.6 versus 25.1). These findings validate that Ita-RC and Ite-RC are complementary, creating a synergistic effect that yields consistent performance gains.

5) Full Integration of PSD, Ita-RC, and Ite-RC: The complete CoreKD framework, which integrates the patch-based semantic contrastive distillation loss ($\mathcal{L}_{psd}$), Ita-RC knowledge constraint ($\mathcal{L}_{ita}$), and Ite-RC knowledge constraint ($\mathcal{L}_{ite}$), achieves the highest mAP of 43.6, marking an overall improvement of 3.4 points from the baseline student model. This comprehensive integration significantly enhances the student detector's performance across all evaluated metrics, as shown in Table V: AP_{50} improves by 3.9 points (62.3 versus 58.4), AP_{75} by 4.5 points (47.8 versus 43.3), AP_S by 4.3 points (26.6 versus 22.3), AP_M by 4.2 points (48.2 versus 44.0), and AP_L by 4.7 points (56.9 versus 52.2). These results highlight the synergistic benefits of combining all components. The integration of local and global contextual information with semantic structural contrastive distillation leads to more robust feature representation and improved learning ability, thereby significantly enhancing the performance of the student.

The ablation studies confirm the individual and combined contributions of each component in the CoreKD framework. Integrating patch-based semantic structural contrastive distillation and contextual knowledge constraints substantially enhances the student detector's learning capabilities, resulting in superior performance across various evaluation metrics.

E. Hyperparameter Analyses

To explore the impact of different hyperparameters on the student model, we have conducted the corresponding experiments on MS COCO 2017, as shown in Fig 3.

1) Sensitivity Study on Patch Size: To investigate the impact of different patch sizes on the student model within the CoreKD framework, we conduct experiments using four distinct patch sizes: 3, 5, 7, and 9, as shown in Fig. 3(a). The results indicate that a patch size of 7 achieves the highest mAP of 43.6, suggesting that this patch size strikes the optimal balance between capturing contextual information and maintaining computational efficiency. As the patch size increases from 3 to 7, there is a noticeable improvement in mAP, which implies that larger patches initially help the model capture more relevant local and contextual features, thereby enhancing detection performance. However, increasing the patch size further to 9 results in a decline in mAP to 43.2. This decrease indicates that excessively large patch sizes may introduce too much background or irrelevant information, which can impede the model's ability to learn effective feature representations.

2) Sensitivity Study on Temperature: To investigate the impact of the temperature coefficient on the student model's performance, we conduct experiments with various values: 0.2, 0.4, 0.5, 0.6, and 0.8, as shown in Fig. 3(b). The results, presented in the figure, reveal that the mAP stabilizes at 43.6 when the temperature coefficient is set to 0.5, 0.6, or 0.8. Notably, there is a slight improvement in mAP as the coefficient increases from 0.2 to 0.5, suggesting that an optimal range for the temperature effectively balances this process.

3) Sensitivity Study on Loss Weight: Considering that our CoreKD framework involves two types of distillation losses ($\mathcal{L}_{psd}$ and $\mathcal{L}_{rcc}$), we further investigate the performance of the student model under different loss weight configurations (λ_1 and λ_2), as shown in Fig. 3(c) and (d).

For λ_1, the optimal values are 0.0001 and 0.0002, both achieving a maximum mAP of 43.6. However, increasing λ_1 to 0.0005 and 0.001 results in a decrease in mAP to 43.5 and 43.3, respectively. This indicates that overly high λ_1 values may negatively affect model performance by possibly overemphasizing the patch-based semantic structural contrastive distillation loss ($\mathcal{L}_{psd}$) at the expense of other important features.

For λ_2, the optimal values are 0.0005 and 0.001, also achieving a mAP of 43.6. Further increases to 0.002 and 0.005 lead to a decrease in mAP to 43.5 and 42.6, respectively. This suggests that an appropriate λ_2 weight enhances model performance by effectively balancing the intra-region ($\mathcal{L}_{ita}$) and inter-region ($\mathcal{L}_{ite}$) contextual knowledge constraints, while excessively high values may introduce too much emphasis on these constraints, resulting in diminishing returns.

Properly balancing the weights of $\mathcal{L}_{psd}$ and $\mathcal{L}_{rcc}$ is crucial for leveraging the full potential of the KD process, ensuring robust feature representation and improved detection accuracy.

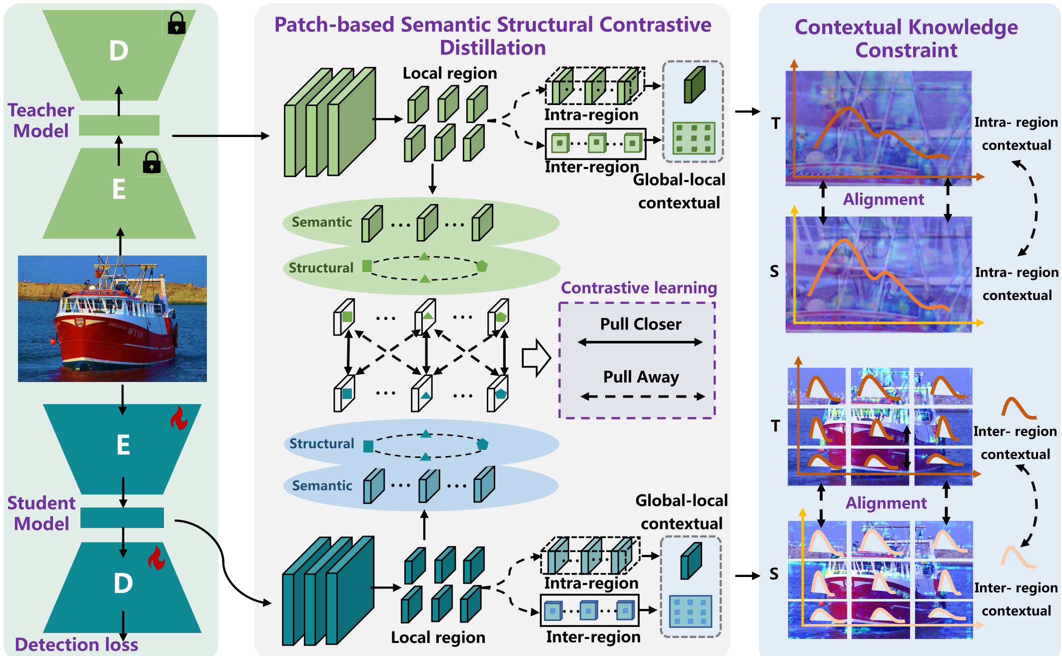

Fig. 2. Overview of the proposed CoreKD framework. First, the intermediate features of the input image are extracted using both the teacher and student detectors. These intermediate features of the feature pyramid network (FPN) [33] are then divided into patches, and a patch-based local region semantic structural contrastive distillation method is employed for consistency and diversity knowledge transfer. In addition, the intra-region and Ite-RC constraints are introduced to further constrain this transfer process, enhancing the student model's contextual modeling capabilities. During the KD process, the parameters of the teacher model remain frozen, while only the parameters of the student model are updated through back-propagation.

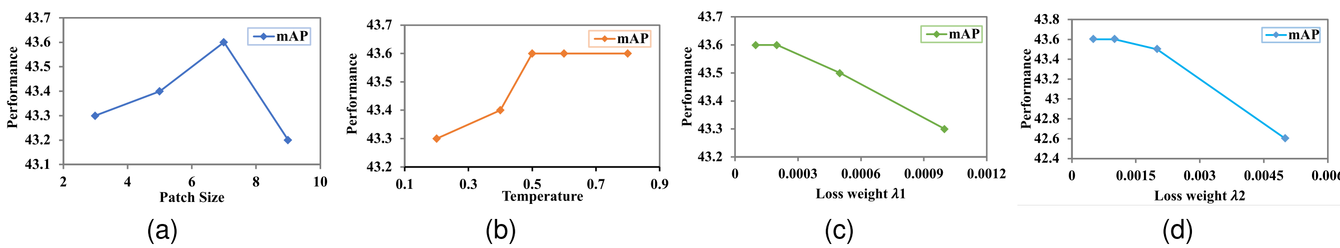

Fig. 3. Hyper-parameter sensitivity study of patch size, τ_s, λ_1, λ_2 with GFL with ResNet-50 backbones on MS COCO2017. (a) Sensitivity study on patch size, (b) sensitivity study on temperature, (c) sensitivity study on loss weight λ_1, and (d) sensitivity study on loss weight λ_2.

TABLE VI

RESULTS OF COREKD FOR LIGHTWEIGHT STUDENT DETECTORS. THE TRAINING SCHEDULE IS ADOPTED FOR 12 EPOCHS IN ALL EXPERIMENTS. THE TEACHER DETECTOR IS GFL WITH RESNET-101 BACKBONES

Backbones	CoreKD	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
MobileNetV2		28.9	45.6	30.4	14.7	31.6	39.2
	✓	32.1 (+3.2)	48.5	34.0	16.7	34.8	43.1
ResNet-18		35.8	53.1	38.2	18.9	38.9	47.9
	✓	38.1 (+2.3)	55.5	41.1	20.6	41.2	51.4
ResNet-34		38.9	56.6	42.2	21.5	42.8	51.4
	✓	41.5 (+2.6)	59.2	45.1	23.4	45.4	54.4
ResNet-50		40.2	58.4	43.3	23.3	44.0	52.2
	✓	43.6 (+3.4)	62.3	47.8	26.6	48.2	56.9

F. Analyses in Different Settings

1) Lightweight Student Detectors: To explore the generalizability of CoreKD, we conduct additional experiments on more lightweight student detectors, as shown in Table VI. We select the GFL [7] detector with a ResNet-101 backbone as the teacher detector. The results demonstrate significant performance improvements across different lightweight models. For the GFL detector with a ResNet-18 backbone, CoreKD increases the mAP by 2.3 points (38.1 versus 35.8), with notable gains in AP_{50}, AP_{75}, AP_S, AP_M, and AP_L. For the ResNet-34 backbone, CoreKD achieves a mAP improvement of 2.6 points (41.5 versus 38.9), again showing consistent gains across all metrics. For the ResNet-50 backbone, the mAP increases by 3.4 points (43.6 versus 40.2), with substantial improvements in AP_{50} (3.9 points), AP_{75} (4.5 points), AP_S (3.3 points), AP_M (4.2 points), and AP_L (4.7 points). These results indicate the CoreKD's effectiveness in enhancing the performance of lightweight student detectors.

We also conduct an experiment under a heterogeneous lightweight distillation setting, where a large-capacity ResNet-101 serves as the teacher and a MobileNetV2 is used as the student. As shown in Table VI, CoreKD improves the baseline performance of MobileNetV2 from 28.9 to 32.1 mAP on the COCO dataset, yielding a significant gain of +3.2 points. This improvement is consistently reflected across all submetrics, including AP_{50} (+2.9), AP_{75} (+3.6), and across small,

TABLE VII

RESULTS OF COREKD ENHANCEMENT WITH MORE POWERFUL TEACHER DETECTORS. THE STUDENT MODELS USE RESNET-50 AS THE BACKBONE NETWORK IN ALL EXPERIMENTS

Teachers	CoreKD	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RetinaNet (T) → RetinaNet (S)							
-	-	37.4	56.7	39.6	20.6	40.7	49.7
ResNet-101	✓	39.8 (+2.4)	58.9	42.6	22.0	43.8	53.5
ResNeXt-101	✓	40.6 (**+3.2**)	60.1	43.1	24.0	44.6	55.5
Mask RCNN (T) → Faster RCNN (S)							
-	-	38.4	59.0	42.0	21.5	42.1	50.3
ResNet-101	✓	40.8 (+2.4)	61.1	44.6	23.0	45.1	54.5
ResNeXt-101	✓	41.5 (**+3.1**)	61.5	45.3	23.4	45.2	55.6

TABLE VIII

COREKD PERFORMANCE IMPROVEMENT ON HETEROGENEOUS TEACHER–STUDENT PAIRS

Methods	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Transformer to CNN						
T: GFL-SwinT	40.5	59.5	43.6	24.0	43.9	53.9
S: GFL-ResNet50	40.2	58.4	43.3	23.3	44.0	52.2
CoreKD	40.8	58.9	44.4	23.4	44.4	53.8
CNN to Transformer						
T: GFL-ResNet101	44.9	63.1	49.0	28.0	49.1	57.2
S: GFL-SwinT	40.5	59.5	43.6	24.0	43.9	53.9
CoreKD	42.0	59.8	45.3	25.3	45.7	55.0

medium, and large objects. These results demonstrate that CoreKD remains effective even in the presence of substantial architectural gaps between the teacher and student.

2) Stronger Teacher Detectors: We further investigate the enhancement of student models by CoreKD under stronger teacher detectors. Experiments are conducted on single-stage (RetinaNet [6]) and two-stage detectors (Faster R-CNN [5]), as shown in Table VII. In these experiments, all student models use ResNet-50 as the backbone network. When using a ResNet-101 RetinaNet as the teacher detector, the student model achieves a mAP of 39.8. However, with a ResNeXt-101 RetinaNet as the teacher detector, the performance further improves to a mAP of 40.6, demonstrating a 3.2 gain over the original student model's mAP of 37.4. In the two-stage detector experiments, utilizing a ResNeXt-101 Mask R-CNN [34] as the teacher detector results in the student's mAP increasing to 41.5, compared to 40.8 with the ResNet-101 Faster R-CNN and 38.4 for the baseline student model.

3) Generalization to Heterogeneous Architectures: We conducted experiments with heterogeneous teacher–student backbone pairs under the COCO benchmark using GFL detectors to further examine the versatility of CoreKD. As shown in Table VIII, when distilling from a transformer teacher (Swin-T) to a CNN student (ResNet-50), CoreKD improves the student's mAP from 40.2 to 40.8. Conversely, when using a CNN teacher (ResNet-101) to guide a transformer student (Swin-T), CoreKD achieves a larger improvement, raising mAP from 40.5 to 42.0 (+1.5) and consistently enhancing AP_{75} (+1.7) and performance across all object scales. These results indicate that CoreKD not only generalizes well across homogeneous CNN backbones but also effectively transfers knowledge between heterogeneous architectures, bridging CNNs and transformers, and further confirming its robustness.

TABLE IX

RESULTS OF COREKD ENHANCEMENT ON THE DETR-BASED DETECTORS

Methods	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
DETR						
T: ResNet50	39.9	60.4	41.7	17.6	43.4	59.4
S: ResNet18	29.0	48.6	29.3	9.8	30.7	46.2
CoreKD	33.7	52.9	34.7	12.4	36.2	53.9
Deformable DETR						
T: ResNet50	47.0	66.1	50.9	30.0	49.8	61.9
S: ResNet18	39.7	57.6	43.2	21.8	42.3	53.8
CoreKD	43.7	62.5	47.2	26.2	46.0	59.2

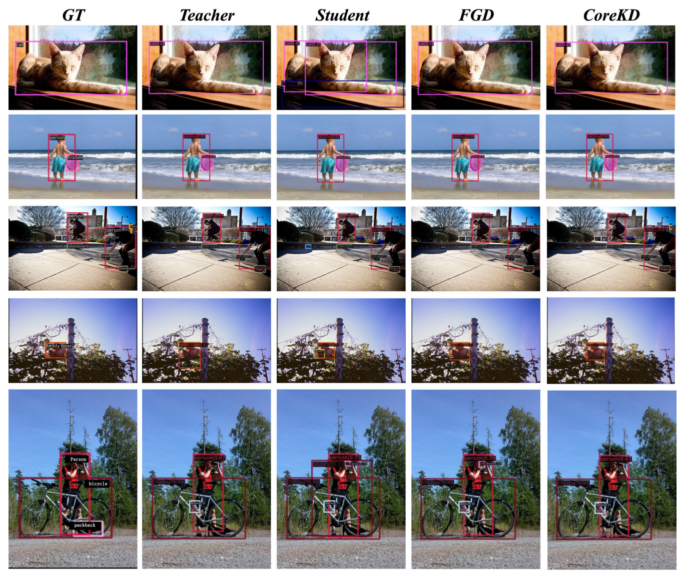

Fig. 4. Visualization of detection results of CoreKD. ***GT***: Ground Truth bounding boxes. ***Teacher***: GFL detector with ResNet-101 backbone. ***Student***: GFL detector with ResNet-50 backbone. ***CoreKD***: student enhanced via the proposed CoreKD framework. ***FGD***: student enhanced via the FGD method.

4) Generalization to DETR-Based Detectors: To further verify the generality of CoreKD, we extend our study to transformer-based detectors, including DETR [73] and deformable DETR [74]. Table IX reports the results with a ResNet-50 teacher and a lightweight ResNet-18 student. CoreKD brings consistent and substantial improvements over the student baselines, achieving +4.7 mAP on DETR and +4.0 mAP on deformable DETR. In particular, CoreKD enhances localization accuracy under stricter IoU thresholds (e.g., +5.4 AP_{75} on DETR), and exhibits strong robustness across object scales, such as +7.7 AP_L and +5.5 AP_M on DETR. These results indicate that the proposed PSD and RCC facilitate effective transfer of structural and contextual knowledge even in transformer-based detection frameworks. Overall, the consistent gains confirm that CoreKD is architecture-agnostic and can serve as a plug-and-play framework across both CNN- and transformer-based detectors.

G. Visualization Analyses

1) Results Visualization Analyses: We visualize the prediction results of the models, as shown in Fig. 4. After introducing our proposed CoreKD, the learning capability of

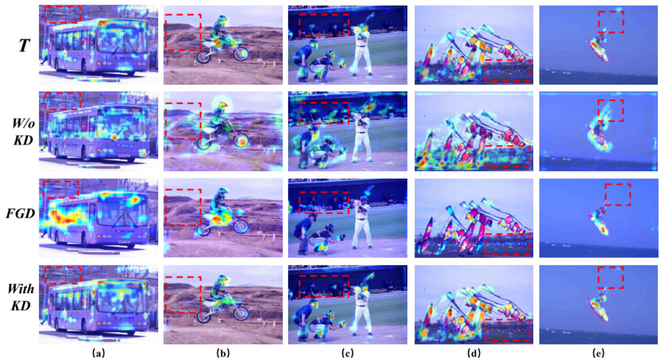

Fig. 5. Heatmap visualizations on the COCO2017 validation set. (a)–(e) Five representative image groups. ***T***: the heatmap from the teacher model. ***w/o KD***: student detector without KD. ***FGD***: student detector distilled with FGD. ***With KD***: student detector distilled with the proposed CoreKD.

the student model is significantly enhanced. For instance, in the first row of the figure, after applying CoreKD, the model can better predict the cat's location and correct an erroneous prediction made by the original student model (where a table was incorrectly predicted as a car). Moreover, we observe that with the introduction of CoreKD, the student model's detection capability can surpass that of the teacher detector in specific scenarios. As illustrated in the third row of Fig. 4, the student model detects a small traffic light that the teacher detector missed. Furthermore, our prediction results have higher confidence and accuracy compared to FGD. These results underscore the effectiveness of CoreKD and its substantial contribution to enhancing the learning ability of the student model.

2) Heatmap Visualization Analyses: Considering that the primary goal of feature-based KD is to enable the student model to learn the feature representation capabilities of the teacher model, we further visualize the heatmaps of both teacher and student models, as shown in Fig. 5. Compared to the original student model's heatmap, the heatmap after introducing our proposed CoreKD is clearer and more focused. This indicates a higher level of attention to foreground objects, which is beneficial for the model to complete the detection tasks. In addition, we observe that the heatmap performance of the student model after introducing CoreKD is similar to that of the teacher model. For instance, in Fig. 5(b), the teacher model's attention is concentrated on the rider's head and the motorcycle's front, while the original student model focuses on the motorcycle's front wheel. After applying CoreKD, the student model's attention aligns closely with that of the teacher model. This demonstrates that our proposed CoreKD effectively enables the student model to learn the feature representations of the teacher model, which can achieve the goal of feature-based knowledge transfer. We observe that while FGD helps the student acquire clean and focused heatmaps, its excessive focus on foreground information may undermine the overall knowledge retained from the teacher model.

H. Discussions

1) Training Efficiency Analysis: Beyond detection performance, we also evaluate the training efficiency of CoreKD in terms of computational cost, as shown in Table X. Compared to FGD [20], CoreKD is more lightweight and faster. Under the same experimental setting (GFL with ResNet-50 teacher and ResNet-18 student on COCO with a 12 schedule), CoreKD consumes less GPU memory (6.25 versus 6.71 GB) and achieves a lower average iteration time (0.5297 versus 0.5991 s/iter). Meanwhile, CoreKD yields a higher mAP (37.1 versus 36.3). This demonstrates that CoreKD provides a favorable balance between effectiveness and efficiency, largely due to its simple design without reliance on additional modules like attention blocks.

TABLE X
TRAINING EFFICIENCY COMPARISON BETWEEN COREKD AND FGD

Method	Manner	GPU Mem. (GB)	Time (s/iter)	mAP
FGD [20]	R50 $\rightarrow$ R18	6.71	0.5991	36.3
CoreKD	R50 $\rightarrow$ R18	6.25	0.5297	37.1

TABLE XI
SENSITIVITY ANALYSIS OF PATCH SIZE ACROSS DATASETS. COREKD REMAINS ROBUST ACROSS DIFFERENT PATCH CONFIGURATIONS

Dataset	3×3	5×5	7×7
MS COCO	43.3	43.4	43.6
PASCAL VOC	59.2	59.3	59.6
VisDrone	15.7	15.8	15.8

2) Robustness to Region Configuration: To evaluate the robustness of CoreKD to the choice of patch size, we conduct a sensitivity analysis on three datasets: MS COCO, PASCAL VOC, and VisDrone. For each dataset, we vary the patch grid sizes (3×3, 5×5, and 7×7) and report the resulting performance in terms of mAP. As shown in Table XI, the variations in detection accuracy are minimal across different patch configurations. This indicates that CoreKD is largely insensitive to the specific patch setting and does not rely on dataset-specific tuning.

3) Limitations: Despite the demonstrated effectiveness of CoreKD across diverse teacher–student configurations, detection architectures, and benchmark datasets, the current design still has certain limitations. In particular, the patch-based local region selection strategy in the PSD module is based on heuristic priors and a fixed granularity. This design choice may constrain the flexibility and generalization of CoreKD in scenarios involving large-scale variations or complex object distributions. Future work will explore adaptive region selection strategies or dynamic patching mechanisms to further enhance robustness and applicability across diverse object detection scenarios.

V. CONCLUSION

This article introduces CoreKD, an innovative KD framework designed for object detection that effectively transfers knowledge from teacher models to student models using a local region contrastive approach. To fully leverage local semantic structural information during the distillation process, we propose a patch-based local region semantic structural contrastive distillation method, enabling the student detector to learn both local consistency and diversity from the teacher model. In addition, to prevent overemphasis on local knowledge, we incorporate Ita-RC and Ite-RC constraints, enhancing the student model's ability to model contextual dependencies. Extensive experiments conducted on the public MS COCO 2017, PASCAL VOC, and VisDrone datasets validate the effectiveness of CoreKD, showcasing its ability to significantly improve the student detectors' object detection performance. We leave the exploration of adaptive region selection and broader architectural extensions as future work.

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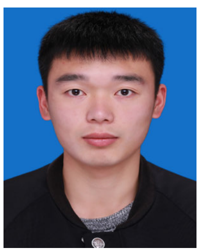

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